Maxime Newman Nereyabagabo

Vancouver, BC | (778) 814-0670 | maximenewman05@gmail.com | LinkedIn | GitHub | Website

Education

Simon Fraser University Expected Graduation: August 2027
Bachelor of Science in Computer Science and Statistics *Burnaby, BC*

- **Relevant Coursework**: Machine Learning, Data Structures and Algorithms, Databases, Operating Systems, Computer Networks

Experience

AI Engineer, Research May 2026 – Present
Tayebi Labs – SFU Cybersecurity

- Co-authoring a **USENIX 2027** paper on an autonomous **multi-agent** pipeline that scans and patches CVEs in **200+ Java and C++** repos
- Generated per-**CVE CodeQL** taint queries from fix commits on **AWS EC2**, verifying **505** of **1,817** traces across **18 CWEs** as patcher input
- Authored per-CVE **PoV** exploits certified against **NVD** and **GHSA**, exposing patches that left vulnerabilities exploitable
- Added **LangGraph** caching, exponential-backoff retries, and idempotency keys, raising unattended pipeline completion

Software Engineer Intern January 2026 – May 2026
SKC Engineering

- Led a **3-person** subteam building an AI-powered **WPS** document generation platform for **ASME Section IX** compliance in **React**/**TypeScript**
- Built an idempotent **Python ETL** pipeline migrating **29,000+** records across **87** tables from **MySQL** to **PostgreSQL** with zero data loss
- Automated dataset curation on **Azure**, processing **15,639** files and eliminating **200+** hours of manual labeling

Software Engineer (AI/ML) Intern September 2025 – January 2026
SKC Engineering

- Built a **FastAPI** backend with **JWT** auth, **PostgreSQL**, and **Redis** caching, cutting p99 latency from **1.2s** to **360ms**
- Engineered **LangGraph** workflows with **RAG** on **Fly Machines** automating **90%** of welding calculations, saving engineers **12+** hours weekly
- Added a **Pytest** and **LangSmith** suite across **60+** cases, cutting workflow regressions **75%** and enabling confident weekly releases

Software Engineer (Systems), Research January 2025 – September 2025
SFU CS Architecture Group

- Reproduced **CUDA GEMM** kernels to **93.7%** of cuBLAS throughput using Nsight Compute profiling and **PTX** assembly analysis
- Researched transformer attention mechanisms, focusing on the large-scale matrix multiplications that dominate **LLM** training and inference
- Authored a **GPU** matrix multiplication kernel guide in **CUDA** on thread blocks, warps, and SM scheduling for undergraduate students

Projects

MapReduce Framework | C++17, Multithreading, CMake GitHub

- Engineered a parallel **MapReduce** framework in **C++17** with mutex-guarded map/shuffle/reduce stages and **multithreading** across workers
- Guaranteed deterministic sorted output independent of thread count, verified across **25** mapper/reducer configurations from **1 to 32** workers
- Built a **CMake** harness with **TSan**/**ASan**/**UBSan** sanitizers, proving thread safety and memory correctness with **zero** data races

FraudLens | Python, Pandas, Scikit-learn, NumPy GitHub | 1st Place – SFU DSSS ML Hackathon

- Built a class-weighted **Random Forest** for multi-class fraud detection, reaching **0.91** macro-F1 on a stratified held-out set
- Engineered **5+** features with leakage checks and stratified splits to boost recall on rare, imbalanced fraud classes
- Shipped an end-to-end training pipeline under hackathon time limits, placing **1st** against **20+** competing teams

Goblin's Keep | TypeScript, Java, JUnit, Mockito GitHub | Play Game

- Refactored a **Java** 2D tile-based escape game into **TypeScript** to run entirely in-browser with no backend dependency
- Implemented **A*** pathfinding with randomized patrol logic to scale difficulty and improve replayability across procedurally generated levels
- Reached **96%** line and **92%** branch coverage with a comprehensive **JUnit** and **Mockito** test suite on the original engine

GreenLeaf Lending Analytics | React, TypeScript, Python GitHub | RBC Hackathon

- Built a **React**/**TypeScript** dashboard scoring loan readiness **0–100** from **25,000+** farm sensor readings across **8** B.C. sites
- Streamed and aggregated **25,000+** sensor readings in real time to surface same-day risk alerts for RBC lenders
- Designed a KPI linking alert latency to crop-stress relief, presented to a panel of RBC judges

Technical Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, Bash
Frameworks & Backend: React, FastAPI, Next.js, Django, REST, gRPC
AI/ML & LLM: LangGraph, Anthropic, Multi-Agent Workflows, RAG, Scikit-learn, Pandas, NumPy, LangSmith
DevOps & Cloud: Docker, AWS, Azure, GCP, GitHub Actions, CI/CD, Redis
Databases: PostgreSQL, MySQL, MongoDB
Testing & Performance: Pytest, JUnit, Mockito, CUDA, Multithreading, TSan/ASan/UBSan
Other: PowerShell, Git, Linux, CodeQL, MCP